

CM 7005 Reality Technologies and Rapid User Experience Design for Communication



Chutisant Kerdvibulvech, Ph.D.
Assistant Professor

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นศ. 7005 เทคโนโลยีเสมือนจริงและการออกแบบประสบการณ์การสื่อสาร

L1

INSTRUCTOR

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(Please Email/Facebook me first to make sure that I will be there.)

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Course Syllabus and Lectures

All course syllabus and lectures can be downloaded online via the following links.

- Course Syllabus
 - <http://gscm.nida.ac.th/~chutisant/CM7005/CS-CM7005.pdf>
- Lectures
 - <http://gscm.nida.ac.th/~chutisant/CM7005/L1.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM7005/L2.pdf>
 - <http://gscm.nida.ac.th/~chutisant/CM7005/L3.pdf>
 - etc

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Evaluation

(Technique-based, Creativity-based, and Research-based)

- Project I / Assignments and Participation
30 %
- Project II / Assignments and Participation (Guest Lecturer)
30%
- Final Project / Assignments and Participation
30%
- Overall Class Participation and Attendance
10 %

Extra Points: 10% for Rinkou

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Textbooks

- Mixed Reality: Merging Real and Virtual Worlds, Yuichi Ohta, Hideyuki Tamura, 418 pages, Springer, ISBN-13: 978-3642875144 (April 15, 2014)
- Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, 2nd Edition (2016)
- สื่อดิจิทัลใหม่...สื่อแห่งอนาคต, ชุดสั้นต์ เกิดวิบูลย์เวช คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์ (นิด้า) 2560

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Related-Academic Papers

- C. Kerdvibulvech, "Vision and Virtual-based Human Computer Interaction Applications for a New Digital Media Visualization", 23rd International Conference on Computer Graphics, Visualization and Computer Vision 2015 (WSCG), Eurographics Association & ACM SIGGRAPH (Scopus, SCImago and Thomson Reuters ISI), Plzen, Czech Republic, June 8 - 12, 2015.
- C. Kerdvibulvech, "An Innovative Use of Multidisciplinary Applications Between Information Technology and Socially Digital Media for Connecting People", The Multidisciplinary International Social Networks Conference 2015 (Springer), in Communication of Computer and Information Science (CCIS), (Scopus and SCImago Quartile 4th), Ehime, Japan, September 1 – 3, 2015.

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Related-Academic Papers

- C. Kerdvibulvech, "A Methodology for Hand and Fingers Motion Analysis Using Adaptive Probabilistic Models," EURASIP Journal on Embedded Systems (JES), 9 pages, No. 18, Springer Publisher (Elsevier's Scopus, EiCompendex, INSPEC, SciFinder, SCImago, ACM, DBLP, and DOAJ), 2014.
- C. Kerdvibulvech and K. Yamauchi, "Structural Human Shape Analysis for Modeling and Recognition," Joint IAPR International Workshops on Structural and Syntactic Pattern Recognition (S+SSPR'14), Lecture Notes in Computer Science (LNCS), vol. 8621, Joensuu, Finland, Springer-Verlag Berlin Heidelberg, pp. 282–290, August 20–22, 2014.
- เอกสารคำสอน นน 7005 การผลิตสื่อดิจิทัล คณะนิเทศศาสตร์และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์ โดย ผศ.ดร.ชุดสั้นต์ เกิดวิบูลย์เวช, 2559.

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Websites and Links

- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/bookseries/558>
- <http://www.dailynews.co.th/>
- <http://www.ecommerce-magazine.com/>
- <http://www.manager.co.th>

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Overview

- Background
- Projects from related-topics of Digital Media

Assoc. Prof. Dr. Chutisant Kerdvibulvech

Education



www.chula.ac.th/

- B.E. (Hons) in Computer Engineering
Chulalongkorn University



www.keio.ac.jp

- MSc. / Ph.D. in Engineering, Department of Information and Computer Science
Keio University, the oldest university in Japan

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Academic Work Experience

- Teaching and being as a research committee in following universities, such as
 - Rangsit University, Sripatum University, PIM
 - Chulalongkorn University, Mahidol University, Kasetsart University
 - NIDA
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

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Media Work Experience

Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



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Digital Media Production

Projects (ภาคปฏิบัติ)

- Introduction (1-2 weeks) + Project I (3 weeks) by Guest Speaker
- Project II (About 4-5 weeks)
 - Touch Communication (National University of Singapore)
 - Chatbot (Artificial Intelligence and Machine Learning)
 - Reality-based Technologies / Facebook 360 / Else
- Final Project (About 4-5 weeks)
 - Website, UX-UI and SEO (Search Engine Optimization)
- Field Trips (About 1-2 weeks)

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Projects (magwileu)

- Introduction (1-2 week)
- Project I (4-5 weeks)
 - Touch Communication (National University of Singapore)
- Project II (About 4-5 weeks)
 - Chatbot (Artificial Intelligence and Machine Learning)
 - Reality-based Technologies / Facebook 360 / Else
- Final Project (About 4-5 weeks)
 - Website, UX/UI and SEO (Search Engine Optimization)
- Field Trips (About 1-2 weeks)

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Digital Media

Digital media are any media that are encoded in a machine-readable format. Digital media can be created, viewed, distributed, modified and preserved on digital electronics devices.

"[Digital Media](#)" (PDF). *Technology Brief*. University of Guelph. September 2006. Retrieved 28 March 2014.

1. Digital Imagery / Digital Video
2. Web Pages / Websites
3. Digital Audio, e.g. mp3s
4. e-books
5. Social Media



aurora.dawn.com

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Future of Digital Media?



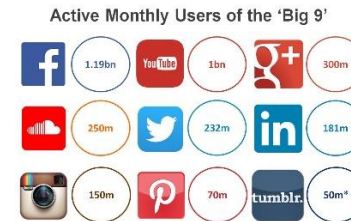
www.3dsg.com.au

- Big Data / SEO (Search Engine Optimization) / SEM (Search Engine Marketing)
- AI (Artificial Intelligence) / Deep Learning
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality) / VR (Virtual Reality)
 - New Digital Media

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Digital Media Trends for 2014

Active monthly users of the 'big 9'



As at 15 November 2013 (multiple sources cited)
*tumblr does not release official monthly user data and figures is based on its estimate from Facebook.com. It is estimated that the actual number is around 50 million per month.

kumber

::Source::

Kamber (Sydney, Australia)

kumber | 14 Social Media Trends for 2014 | Assoc. Prof. Dr. Chutisant Kerdvibulvech | Kamber.com.au | 14

Digital Media Trends for 2016



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<https://www.facebook.com/zuck>

Digital Media Trends for 2017/2018



Facebook, Inc. Common Stock
NASDAQ: FB - Mar 28, 6:15 AM EDT

152.22 USD $\uparrow$ 7.84 (4.90%) Facebook's share price peak

After-hours: 151.38 $\uparrow$ 0.55%

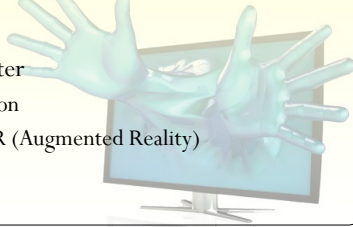


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<https://www.verdict.co.uk/facebook-has-now-lost-100-billion-in-market-cap-since-february/>

The Future of Digital Media

- Big Data / Cloud Computing / SEO (Search Engine Optimization) – SEM (Search Engine Marketing)
- From Mobile-Oriented to AI-Oriented
- Reaching-Out-of-Computer
 - 3D-Display / 3D Television
 - MR (Mixed Reality) / AR (Augmented Reality)



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Augmented Reality vs. Virtual Reality

Augmented Reality

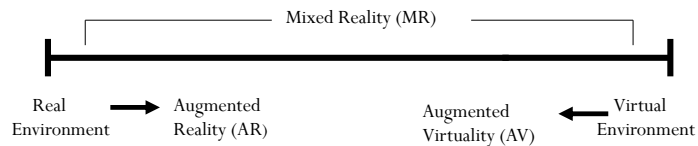
- System augments the real world scene
- User maintains a sense of presence in real world
- Needs a mechanism to combine virtual and real worlds

Virtual Reality:

- Totally immersive environment
- Visual senses are under control of system

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Miligram's Reality-Virtuality Continuum



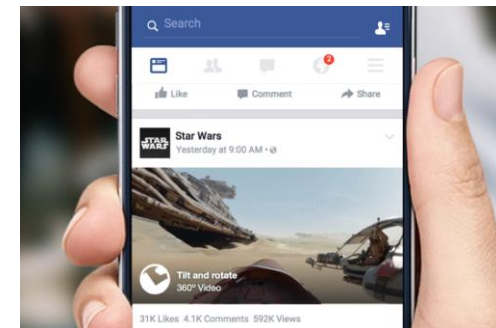
csis.pace.edu

Miligram coined the term “AugmentedVirtuality” to identify systems which are mostly synthetic with some real world imagery added such as texture mapping video onto virtual objects.

Augmented Reality is closest to the real world because mainly a user is perceiving the real world with just a little computer generated data.

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From Virtual Reality to Facebook360



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Virtual Reality



“The Oculus community is leading us into the future, and we can’t wait for everyone to experience the magic of VR and the next platform.”...**Mark Zuckerberg, CEO of Facebook**

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24 September 2015

Chatbot

- A computer program which conducts a conversation via auditory or textual methods
- A chatter robot is a type of conversational agent, a computer program designed to simulate an intelligent conversation with one or more human users via auditory or textual methods
- AI (Artificial Intelligence) / Deep Learning

<https://en.wikipedia.org/wiki/Chatterbot>

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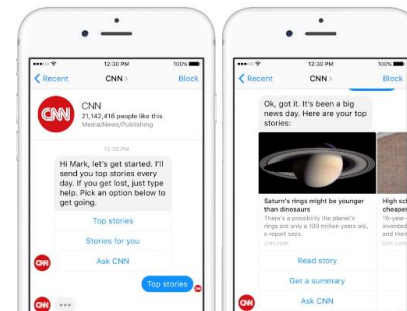
When Facebook understands our messages automatically..

เมื่อเฟซบุ๊กเข้าใจข้อความของเราได้เอง (1) (2) (3)..หนังสือพิมพ์เดลินิวส์

- <http://www.dailynews.co.th/it/502607>
- <http://repository.nida.ac.th/bitstream/handle/662723737/3148/59104.pdf?sequence=1&isAllowed=y>
- <http://repository.nida.ac.th/handle/662723737/3151>

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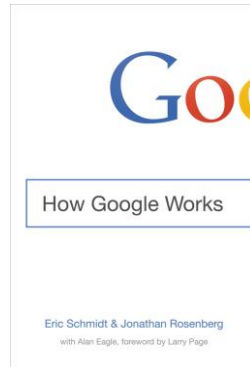
Chatbot by CNN



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Rinkou: Extra Points 10%

- <https://www.amazon.com/How-Google-Works-Eric-Schmidt/dp/1455582344>



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Thank you

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